LAB 3 — ARCHITECTURE JUSTIFICATION

Municipal Solid Waste Route Optimizer

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Architecture Selection

We chose Layered Architecture for the Municipal Solid Waste Route Optimizer.

Architectural Choice & Reasons

1. Clear separation of responsibilities: The Presentation Layer handles the municipal administrator and driver interface, the Business Layer handles collection management and route optimization, and the Data / Integration Layer manages waste and vehicle data together with GPS / Map communication. This separation makes the system easier to understand, test, and maintain.

2. Suitable for the system workflow: Requests can move from the user interface to Collection Management and then to the Route Optimization Engine, while required information is obtained through the Data / Integration Layer. This supports managing collection points and vehicles, generating optimized routes, and allowing drivers to view assigned routes and update collection status.

Security Advantage

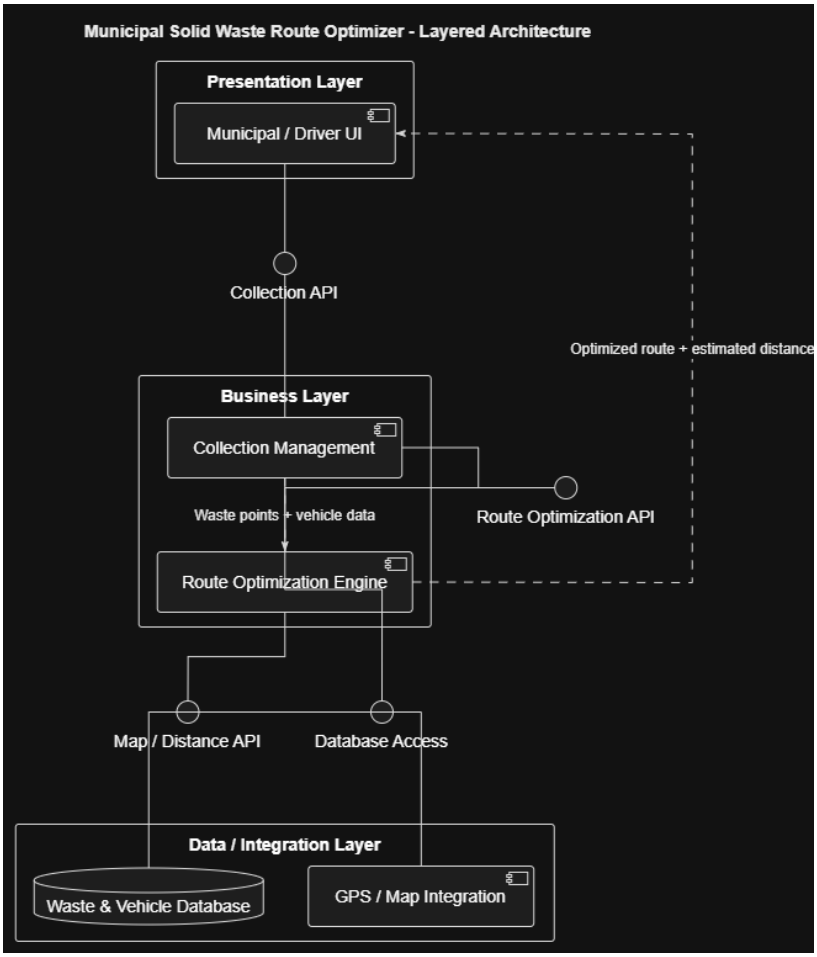
The layered structure separates the user interface from the database. Database access can be restricted to the Data / Integration Layer, preventing municipal users and drivers from directly accessing stored waste and vehicle records. Authentication and authorization can be enforced at the application boundary.

Performance Benefit

The Route Optimization Engine is kept as a dedicated business component, allowing route-generation processing to be optimized independently of the user interface. Waste-point and vehicle information can also be retrieved through the data layer, helping keep the interface responsive during route generation.

Components Used in the Component Diagram

1. Municipal / Driver UI 2. Collection Management 3. Route Optimization Engine 4. Waste & Vehicle Database 5. GPS / Map Integration


Municipal Solid Waste Route Optimizer - Layered Architecture
Presentation Layer
Municipal / Driver UI
Collection API
Optimized route + estimated distance
Business Layer
Collection Management
Waste points + vehicle data
Route Optimization API
Route Optimization Engine
Map / Distance API
Database Access
Data / Integration Layer
Waste & Vehicle Database
GPS / Map Integration